Workshop Example

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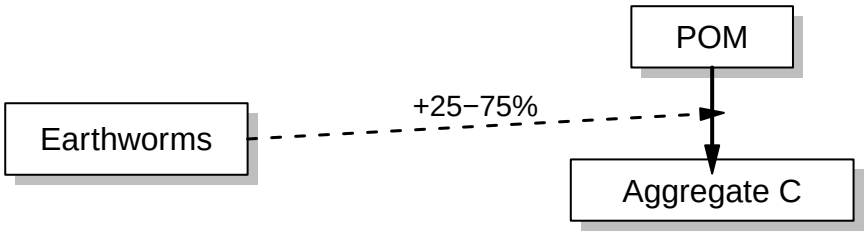
Rationale

This is a small example of how I proposed that we proceed with our analysis. I am showing just a simple example of one possible indirect effect, but the goal is to be more comprehensive. I am using the Millennial Model as an example.

Here is the example of earthworms affecting aggregation rate of POM into soil aggregates. Our literature search suggests that earthworms might increase this rate by somewhere between 25 and 75% (I just made this up for this example).

```
## Loading required package: shape
```



Lever

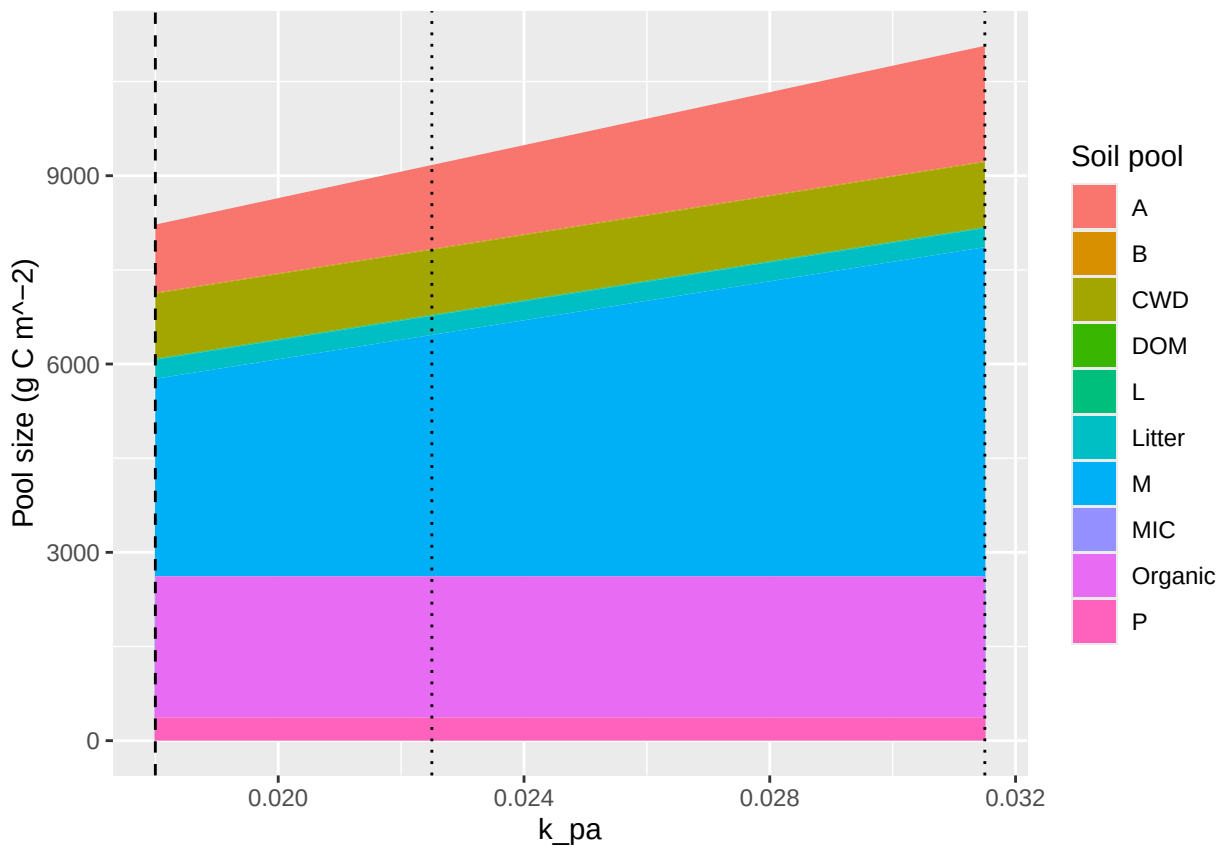
In the Millennial Model, the aggregation rate is controlled by the parameter `k_pa`, so this is the one that we think earthworms will modify. The first, basic option is to test how much changing this parameter might

influence carbon distribution in the soil pools.

Testing the lever

```
source("Scripts/01a_Millennial_ode.R")
source("Scripts/01b_Millennial_parameters.R")
## - Units: All fluxes in g C m^-2 d^-1; states in g C m^-2; moisture in m^3 m^-3; temperature in ˚C.

y0       <- c(Litter  = 200,
              CWD     = 1000,
              Organic = 2000,
              DOM = 10,
              MIC = 0.1,
              P   = 400,
              L   = 10,
              A   = 1000,
              M   = 3000,
              B   = 40)


# Run model to equilibrium:
parms["force_eqm"] = 1

# Calculate the equilibria across parameter ranges:

sens <- data.frame(
  Parameter = c("k_pa"),
  Baseline = c(0.018),
  High = c(0.018*1.75),
  Low = c(0.018*1.25)
)

# Millennial model sensitivity:

results <- vector(mode = "list", 1)
names(results) = sens$Parameter

BASELINE = rootSolve::stode(y = y0,
                     func = millennial_v2_ode,
                     parms = parms)$y

for(curpar in 1){
  pargrad <- seq(sens[curpar,4], sens[curpar,3], length = 10)

  results[[curpar]] <- t(sapply(pargrad, FUN = function(X, nnn = curpar){
    pcur = parms
    pcur[sens[nnn,1]] = X
    rootSolve::stode(y = y0,
                     func = millennial_v2_ode,
                     parms = pcur)$y
  }))

  results[[curpar]] = as.data.frame(results[[curpar]])
```

```
  results[[curpar]] <- cbind(results[[curpar]], par = pargrad, parname = names(results)[[curpar]])
}

results$k_pa %>%
  pivot_longer(!par & !parname) %>%
  bind_rows(
    tibble(par = 0.018, parname = "k_pa", name = names(BASELINE), value = BASELINE)
  ) %>%
  ggplot(aes(x = par, y = value, fill = name)) +
  geom_area() +
  geom_vline(xintercept = 0.018, linetype = 2) +
  geom_vline(xintercept = c(0.0225, 0.0315), linetype = 3) +
  ylab("Pool size (g C m^-2)") + xlab("k_pa") + scale_fill_discrete(name = "Soil pool")
```

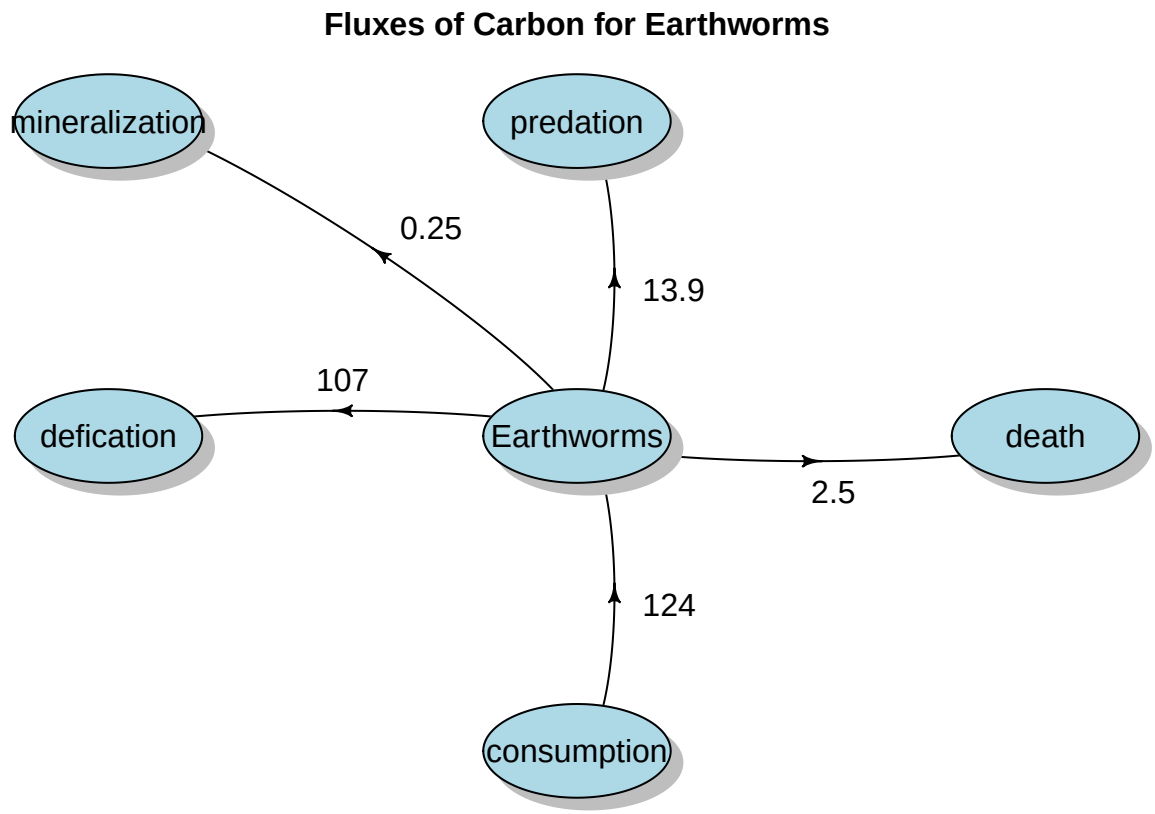


In the above figure, you can see the baseline model at the dashed line on the left. The dotted lines bound the model predictions with the increased aggregate formation rate. Increasing this rate, increases the amount of aggregated carbon (A) and the amount of mineral-associated carbon (M). These changes amount to an equilibrium increase of at least 200 g C m^-2, which is a lot (maybe not realistic...).

To decide how relevant this is to include in a coupled, let's go to the soil food web model and consider the effects of these earthworms on direct carbon cycling. If the earthworm direct effect on the same order of magnitude or less than this indirect effect, we probably need to include this into our coupled model to accurately estimate the earthworm effect.

```
library(multnutFW)
```

```r
# Build a simple community as an example using the default community in the package:
sce = intro_comm

# Rename the prey2 detritivore node for clarity:
sce = renamenode(sce, "Prey2", "Earthworms")

# Change the feeding to match:
sce$imat[3,] = c(0,0,0,0.1, 0.9,0)
# Change assimilation to be more realistic:
sce$prop$assimilation$Carbon[3,5] = 0.1

node_flux_view(sce, node = "Earthworms", Element = "Carbon")
```



So in this system (again numbers currently made up), earthworms are consuming ~124 g C m^-2 from detritus and microbes and causing around 0.25 of that to be released as CO2. Lots of this carbon that they consume (107) gets returned to the organic soil pools because of their low assimilation efficiency.

Overall, their direct effects of moving this carbon are large, but the large change we saw in the stocks caused by *stabilizing* that carbon, probably means it is an important mechanism to include.

With this conclusion, we can begin to justify building the above mechanism (earthworms increase `k_pa` into a coupled model).

Note: Code available here: https://github.com/robertwbuchkowski/merging_SFW_decomposition_models